Customer Load Violation Model

Current Findings

21 September 2026

1 Mathematical formulation

N customers are divided into small (S), medium (M), and large (L) classes. Let n_S, n_M, n_L denote the number of customers in each class and let I_S, I_M, I_L denote their respective current demands. This means:

$$n_S + n_M + n_L = N, \qquad n_S, n_M, n_L \in \{0, 1, \ldots, N\}. \tag{1}$$

Each customer independently belongs to class S, M or L with probabilities p_S, p_M, p_L, where $p_S + p_M + p_L = 1$ and $p_j \geq 0$. The random class counts follow a trinomial distribution:

$$(N_S, N_M, N_L) \sim \text{Trinomial}(N; p_S, p_M, p_L). \tag{2}$$

For a composition $\boldsymbol{n} = (n_S, n_M, n_L)$, its probability is

$$w(\boldsymbol{n}) = \mathbb{P}(N_S = n_S, N_M = n_M, N_L = n_L) = \frac{N!}{n_S! n_M! n_L!} p_S^{n_S} p_M^{n_M} p_L^{n_L}. \tag{3}$$

And for customer i in class $j \in \{S, M, L\}$, define the state $X_{j,i} \in \{0, 1\}$, where 1 means on and 0 means off. The aggregate current is:

$$A = \sum_{j \in \{S,M,L\}} I_j \sum_{i=1}^{n_j} X_{j,i}. \tag{4}$$

A violation occurs when the aggregate current is strictly above the network limit T. Its probability is:

$$\mathbb{P}(A > T). \tag{5}$$

The current study uses:

$$(I_S, I_M, I_L) = (20, 40, 100), \qquad N = 40, \qquad T = 1000. \tag{6}$$

2 All-On model

Assuming all customer loads are on, every state variable is fixed at $X_{j,i} = 1$, so the aggregate current for a given composition is:

$$A_{\text{all}} = I_S n_S + I_M n_M + I_L n_L. \tag{7}$$

Consequently, the violation probability for a fixed composition is

$$\mathbb{P}(A > T \mid n_S, n_M, n_L, \text{all on}) = \begin{cases} 1, & I_S n_S + I_M n_M + I_L n_L > T, \\ 0, & I_S n_S + I_M n_M + I_L n_L \leq T. \end{cases} \tag{8}$$

The graph demonstrates the violating and non-violating composition points for the parameters $I_S = 20$, $I_M = 40$, $I_L = 100$, $N = 40$, and $T = 1000$.

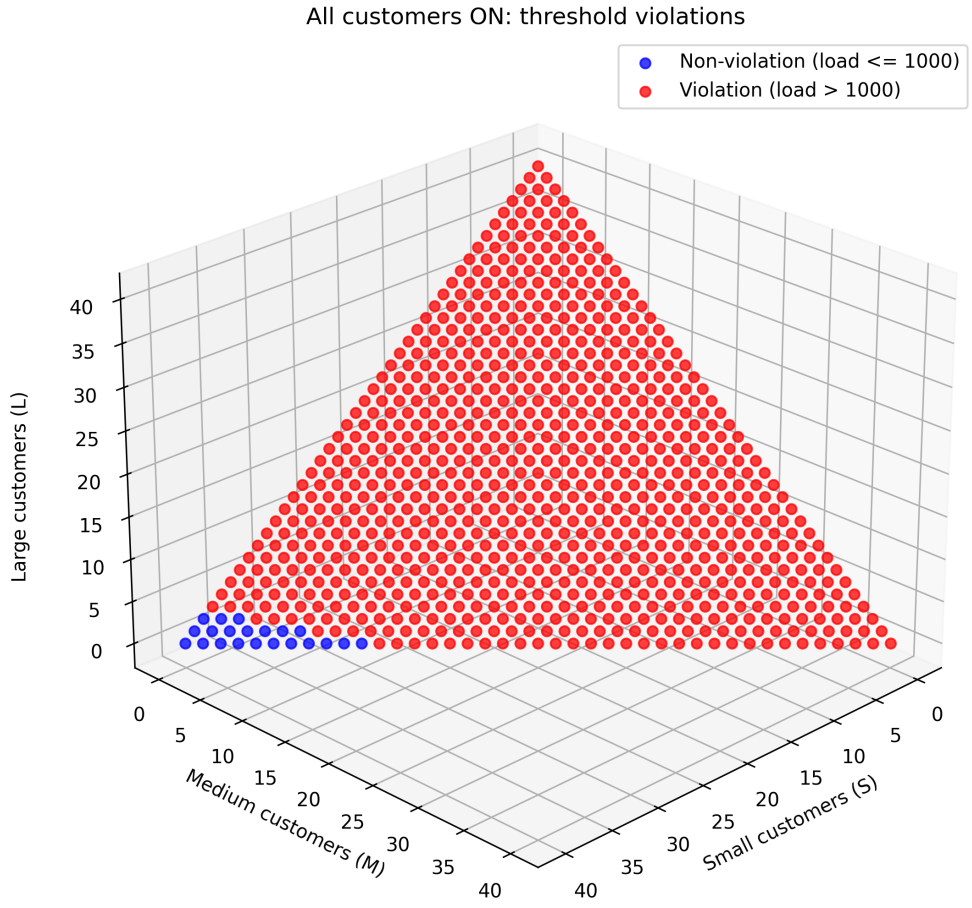


Figure 1: All-On model results.

To count the total number of compositions, each composition corresponds to a different placement of the two separators in the array:

$$\underbrace{\star\cdots\star}_{n_S} \;\Big|\; \underbrace{\star\cdots\star}_{n_M} \;\Big|\; \underbrace{\star\cdots\star}_{n_L} \quad .$$

Choosing the two separator positions among $N + 2$ positions gives

$$\text{Number of compositions} = \binom{N+2}{2} = \frac{(N+2)(N+1)}{2}. \tag{9}$$

3 On–Off model

For a fixed composition (n_S, n_M, n_L), let q_S, q_M, q_L be the respective probabilities that a customer is on. Customers switch independently, so the numbers of active customers are indepen-

dent binomial random variables:

$$Y_S \sim \text{Binomial}(n_S, q_S), \quad Y_M \sim \text{Binomial}(n_M, q_M), \quad Y_L \sim \text{Binomial}(n_L, q_L). \tag{10}$$

Here n_j is the fixed number of customers in class j, whereas $Y_j \in \{0, \ldots, n_j\}$ is the random number that are on. The aggregate current is

$$A = I_S Y_S + I_M Y_M + I_L Y_L. \tag{11}$$

The probability of a violation for this composition is therefore

$$\mathbb{P}(A > T \mid n_S, n_M, n_L) = \mathbb{P}(I_S Y_S + I_M Y_M + I_L Y_L > T). \tag{12}$$

Using independence, this probability can be calculated exactly as

$$\mathbb{P}(A > T \mid n_S, n_M, n_L) = \sum_{\substack{y_S=0,\ldots,n_S;\ y_M=0,\ldots,n_M;\ y_L=0,\ldots,n_L \\ I_S y_S + I_M y_M + I_L y_L > T}} \prod_{j \in \{S,M,L\}} \binom{n_j}{y_j} q_j^{y_j} (1-q_j)^{n_j - y_j}. \tag{13}$$

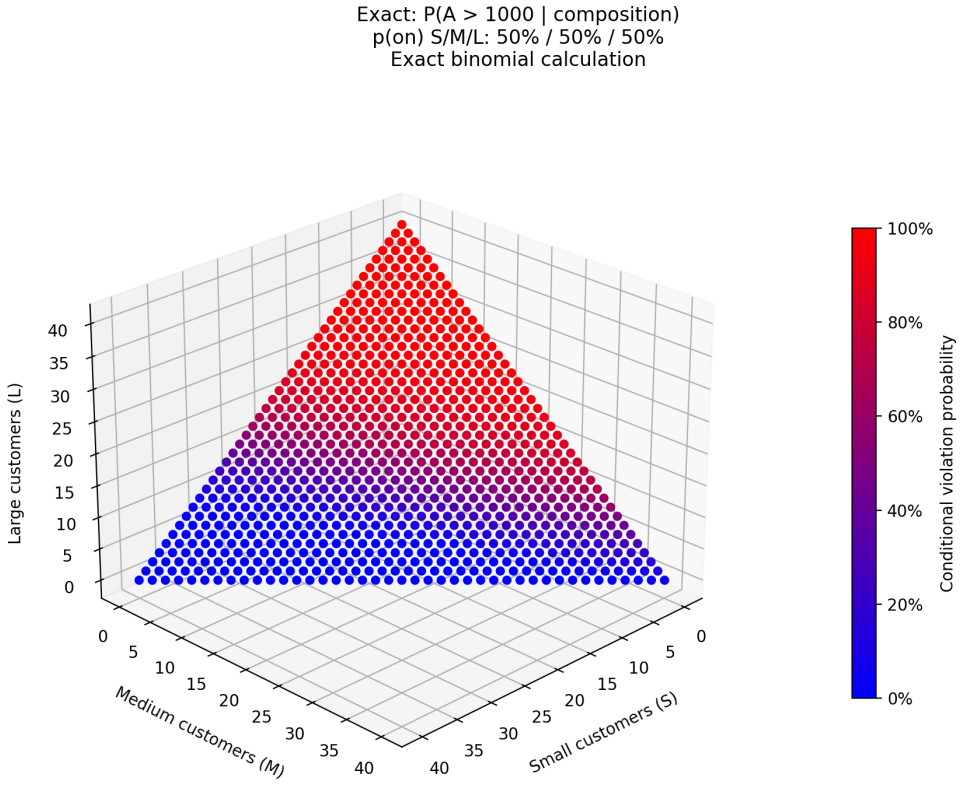


Figure 2: Exact conditional violation probabilities at $q_S = q_M = q_L = 0.5$, calculated using binomial probabilities without sampling.

For a population with customer type probabilities p_S, p_M, p_L, the overall violation probability is

$$\mathbb{P}(A > T) = \sum_{\substack{n_S, n_M, n_L \geq 0 \\ n_S + n_M + n_L = N}} w(\boldsymbol{n})\, \mathbb{P}(A > T \mid n_S, n_M, n_L). \tag{14}$$